

Thermodynamics

Instructor

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Chapter 10 – Problems

**Problem numbers correspond to
5th Edition**

10–4 A steady-flow Carnot cycle uses water as the working fluid. Water changes from saturated liquid to saturated vapor as heat is transferred to it from a source at 250°C . Heat rejection takes place at a pressure of 20 kPa. Show the cycle on a T - s diagram relative to the saturation lines, and determine (a) the thermal efficiency, (b) the amount of heat rejected, in kJ/kg, and (c) the net work output.

Analysis (a) Noting that $T_H = 250^\circ\text{C} = 523\text{ K}$ and $T_L = T_{\text{sat @ } 20\text{ kPa}} = 60.06^\circ\text{C} = 333.1\text{ K}$, the thermal efficiency becomes

$$\eta_{\text{th,C}} = 1 - \frac{T_L}{T_H} = 1 - \frac{333.1\text{ K}}{523\text{ K}} = 0.3632 = \mathbf{36.3\%}$$

(b) The heat supplied during this cycle is simply the enthalpy of vaporization,

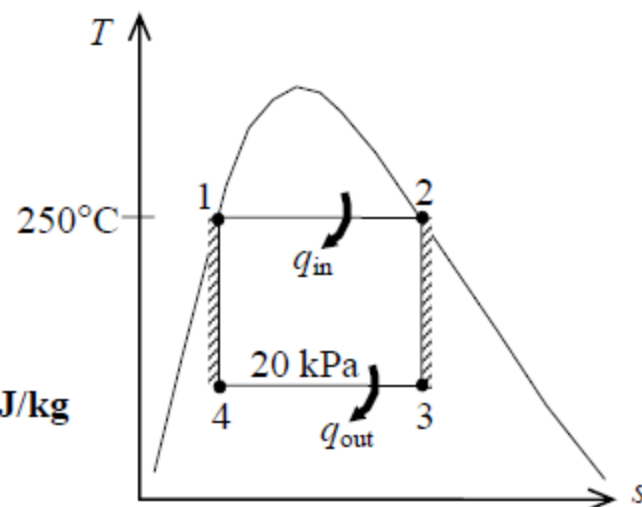
$$q_{\text{in}} = h_{fg@250^\circ\text{C}} = 1715.3\text{ kJ/kg}$$

Thus,

$$q_{\text{out}} = q_L = \frac{T_L}{T_H} q_{\text{in}} = \left(\frac{333.1\text{ K}}{523\text{ K}} \right) (1715.3\text{ kJ/kg}) = \mathbf{1092.3\text{ kJ/kg}}$$

(c) The net work output of this cycle is

$$w_{\text{net}} = \eta_{\text{th}} q_{\text{in}} = (0.3632)(1715.3\text{ kJ/kg}) = \mathbf{623.0\text{ kJ/kg}}$$



10-5 Repeat Prob. 10-4 for a heat rejection pressure of 10 kPa.

10-6 Consider a steady-flow Carnot cycle with water as the working fluid. The maximum and minimum temperatures in the cycle are 350 and 60°C. The quality of water is 0.891 at the beginning of the heat-rejection process and 0.1 at the end. Show the cycle on a T - s diagram relative to the saturation lines, and determine (a) the thermal efficiency, (b) the pressure at the turbine inlet, and (c) the net work output.

Answers: (a) 0.465, (b) 1.40 MPa, (c) 1623 kJ/kg

The Simple Rankine Cycle

10–20 Consider a coal-fired steam power plant that produces 300 MW of electric power. The power plant operates on a simple ideal Rankine cycle with turbine inlet conditions of 5 MPa and 450°C and a condenser pressure of 25 kPa. The coal has a heating value (energy released when the fuel is burned) of 29,300 kJ/kg. Assuming that 75 percent of this energy is transferred to the steam in the boiler and that the electric generator has an efficiency of 96 percent, determine (a) the overall plant efficiency (the ratio of net electric power output to the energy input as fuel) and (b) the required rate of coal supply. *Answers: (a) 24.5 percent, (b) 150 t/h*

Analysis (a) From the steam tables (Tables A-4, A-5, and A-6),

$$h_1 = h_f @ 25 \text{ kPa} = 271.96 \text{ kJ/kg}$$

$$\nu_1 = \nu_f @ 25 \text{ kPa} = 0.001020 \text{ m}^3/\text{kg}$$

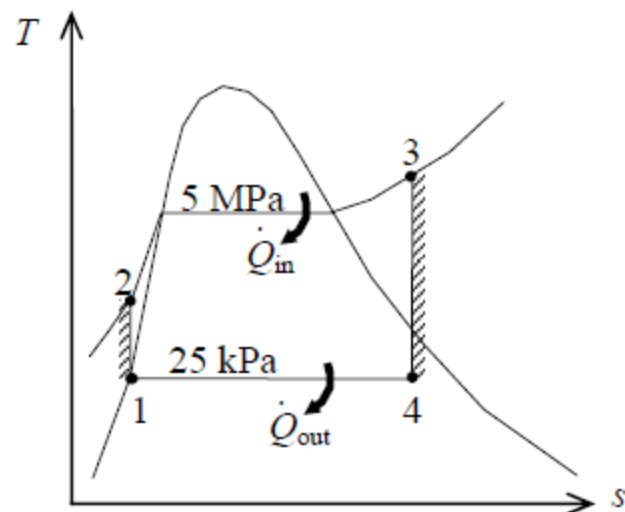
$$\begin{aligned} w_{p,\text{in}} &= \nu_1 (P_2 - P_1) \\ &= (0.00102 \text{ m}^3/\text{kg})(5000 - 25 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 5.07 \text{ kJ/kg} \end{aligned}$$

$$h_2 = h_1 + w_{p,\text{in}} = 271.96 + 5.07 = 277.03 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_3 = 5 \text{ MPa} \\ T_3 = 450^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 = 3317.2 \text{ kJ/kg} \\ s_3 = 6.8210 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_4 = 25 \text{ kPa} \\ s_4 = s_3 \end{array} \right\} x_4 = \frac{s_4 - s_f}{s_{fg}} = \frac{6.8210 - 0.8932}{6.9370} = 0.8545$$

$$h_4 = h_f + x_4 h_{fg} = 271.96 + (0.8545)(2345.5) = 2276.2 \text{ kJ/kg}$$



The thermal efficiency is determined from

$$q_{\text{in}} = h_3 - h_2 = 3317.2 - 277.03 = 3040.2 \text{ kJ/kg}$$
$$q_{\text{out}} = h_4 - h_1 = 2276.2 - 271.96 = 2004.2 \text{ kJ/kg}$$

and

$$\eta_{\text{th}} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{2004.2}{3040.2} = 0.3407$$

Thus,

$$\eta_{\text{overall}} = \eta_{\text{th}} \times \eta_{\text{comb}} \times \eta_{\text{gen}} = (0.3407)(0.75)(0.96) = \mathbf{24.5\%}$$

(b) Then the required rate of coal supply becomes

$$\dot{Q}_{\text{in}} = \frac{\dot{W}_{\text{net}}}{\eta_{\text{overall}}} = \frac{300,000 \text{ kJ/s}}{0.2453} = 1,222,992 \text{ kJ/s}$$

and

$$\dot{m}_{\text{coal}} = \frac{\dot{Q}_{\text{in}}}{C_{\text{coal}}} = \frac{1,222,992 \text{ kJ/s}}{29,300 \text{ kJ/kg}} \left(\frac{1 \text{ ton}}{1000 \text{ kg}} \right) = 0.04174 \text{ tons/s} = \mathbf{150.3 \text{ tons/h}}$$

10–21 Consider a solar-pond power plant that operates on a simple ideal Rankine cycle with refrigerant-134a as the working fluid. The refrigerant enters the turbine as a saturated vapor at 1.4 MPa and leaves at 0.7 MPa. The mass flow rate of the refrigerant is 3 kg/s. Show the cycle on a T - s diagram with respect to saturation lines, and determine (a) the thermal efficiency of the cycle and (b) the power output of this plant.

10–22 Consider a steam power plant that operates on a simple ideal Rankine cycle and has a net power output of 45 MW. Steam enters the turbine at 7 MPa and 500°C and is cooled in the condenser at a pressure of 10 kPa by running cooling water from a lake through the tubes of the condenser at a rate of 2000 kg/s. Show the cycle on a T - s diagram with respect to saturation lines, and determine (a) the thermal efficiency of the cycle, (b) the mass flow rate of the steam, and (c) the temperature rise of the cooling water. *Answers: (a) 38.9 percent, (b) 36 kg/s, (c) 8.4°C*

Analysis (a) From the steam tables (Tables A-4, A-5, and A-6),

$$h_1 = h_f @ 10 \text{ kPa} = 191.81 \text{ kJ/kg}$$

$$v_1 = v_f @ 10 \text{ kPa} = 0.00101 \text{ m}^3/\text{kg}$$

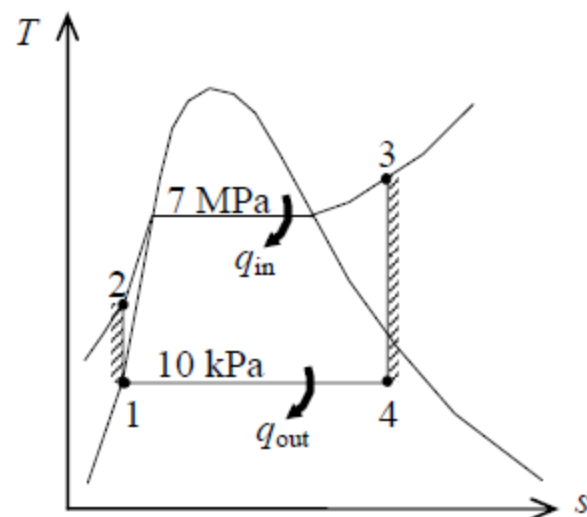
$$\begin{aligned} w_{p,\text{in}} &= v_1(P_2 - P_1) \\ &= (0.00101 \text{ m}^3/\text{kg})(7,000 - 10 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 7.06 \text{ kJ/kg} \end{aligned}$$

$$h_2 = h_1 + w_{p,\text{in}} = 191.81 + 7.06 = 198.87 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_3 = 7 \text{ MPa} \\ T_3 = 500^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 = 3411.4 \text{ kJ/kg} \\ s_3 = 6.8000 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_4 = 10 \text{ kPa} \\ s_4 = s_3 \end{array} \right\} x_4 = \frac{s_4 - s_f}{s_{fg}} = \frac{6.8000 - 0.6492}{7.4996} = 0.8201$$

$$h_4 = h_f + x_4 h_{fg} = 191.81 + (0.8201)(2392.1) = 2153.6 \text{ kJ/kg}$$



Thus,

$$q_{\text{in}} = h_3 - h_2 = 3411.4 - 198.87 = 3212.5 \text{ kJ/kg}$$

$$q_{\text{out}} = h_4 - h_1 = 2153.6 - 191.81 = 1961.8 \text{ kJ/kg}$$

$$w_{\text{net}} = q_{\text{in}} - q_{\text{out}} = 3212.5 - 1961.8 = 1250.7 \text{ kJ/kg}$$

and

$$\eta_{\text{th}} = \frac{w_{\text{net}}}{q_{\text{in}}} = \frac{1250.7 \text{ kJ/kg}}{3212.5 \text{ kJ/kg}} = \mathbf{38.9\%}$$

$$(b) \quad \dot{m} = \frac{\dot{W}_{\text{net}}}{w_{\text{net}}} = \frac{45,000 \text{ kJ/s}}{1250.7 \text{ kJ/kg}} = \mathbf{36.0 \text{ kg/s}}$$

(c) The rate of heat rejection to the cooling water and its temperature rise are

$$\dot{Q}_{\text{out}} = \dot{m}q_{\text{out}} = (35.98 \text{ kg/s})(1961.8 \text{ kJ/kg}) = 70,586 \text{ kJ/s}$$

$$\Delta T_{\text{cooling water}} = \frac{\dot{Q}_{\text{out}}}{(\dot{m}c)_{\text{cooling water}}} = \frac{70,586 \text{ kJ/s}}{(2000 \text{ kg/s})(4.18 \text{ kJ/kg} \cdot ^\circ\text{C})} = \mathbf{8.4^\circ\text{C}}$$

10-23 Repeat Prob. 10-22 assuming an isentropic efficiency of 87 percent for both the turbine and the pump.

Answers: (a) 33.8 percent, (b) 41.4 kg/s, (c) 10.5°C

Analysis (a) From the steam tables (Tables A-4, A-5, and A-6),

$$h_1 = h_f @ 10 \text{ kPa} = 191.81 \text{ kJ/kg}$$

$$\nu_1 = \nu_f @ 10 \text{ kPa} = 0.00101 \text{ m}^3/\text{kg}$$

$$\begin{aligned} w_{p,\text{in}} &= \nu_1(P_2 - P_1) / \eta_p \\ &= (0.00101 \text{ m}^3/\text{kg})(7,000 - 10 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) / (0.87) \\ &= 8.11 \text{ kJ/kg} \end{aligned}$$

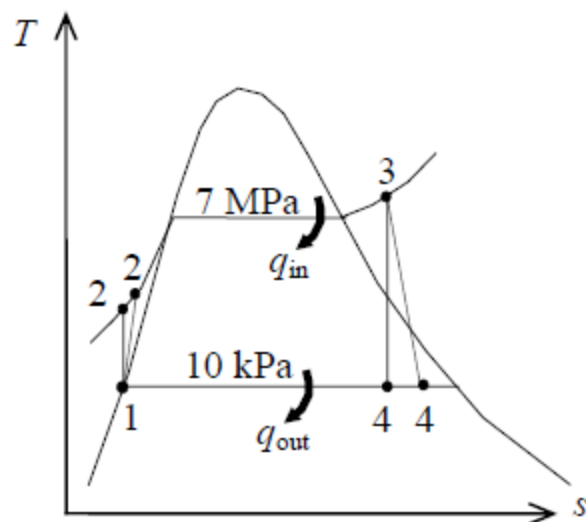
$$h_2 = h_1 + w_{p,\text{in}} = 191.81 + 8.11 = 199.92 \text{ kJ/kg}$$

$$\begin{aligned} P_3 &= 7 \text{ MPa} \left\{ \begin{aligned} h_3 &= 3411.4 \text{ kJ/kg} \\ T_3 &= 500^\circ\text{C} \end{aligned} \right. \left\{ \begin{aligned} s_3 &= 6.8000 \text{ kJ/kg} \cdot \text{K} \end{aligned} \right. \end{aligned}$$

$$\begin{aligned} P_4 &= 10 \text{ kPa} \\ s_4 &= s_3 \end{aligned} \left\{ \begin{aligned} x_4 &= \frac{s_4 - s_f}{s_{fg}} = \frac{6.8000 - 0.6492}{7.4996} = 0.8201 \end{aligned} \right.$$

$$h_{4s} = h_f + x_4 h_{fg} = 191.81 + (0.8201)(2392.1) = 2153.6 \text{ kJ/kg}$$

$$\begin{aligned} \eta_T &= \frac{h_3 - h_4}{h_3 - h_{4s}} \longrightarrow h_4 = h_3 - \eta_T(h_3 - h_{4s}) \\ &= 3411.4 - (0.87)(3411.4 - 2153.6) = 2317.1 \text{ kJ/kg} \end{aligned}$$



Thus,

$$q_{\text{in}} = h_3 - h_2 = 3411.4 - 199.92 = 3211.5 \text{ kJ/kg}$$

$$q_{\text{out}} = h_4 - h_1 = 2317.1 - 191.81 = 2125.3 \text{ kJ/kg}$$

$$w_{\text{net}} = q_{\text{in}} - q_{\text{out}} = 3211.5 - 2125.3 = 1086.2 \text{ kJ/kg}$$

and

$$\eta_{\text{th}} = \frac{w_{\text{net}}}{q_{\text{in}}} = \frac{1086.2 \text{ kJ/kg}}{3211.5 \text{ kJ/kg}} = \mathbf{33.8\%}$$

$$(b) \quad \dot{m} = \frac{\dot{W}_{\text{net}}}{w_{\text{net}}} = \frac{45,000 \text{ kJ/s}}{1086.2 \text{ kJ/kg}} = \mathbf{41.43 \text{ kg/s}}$$

(c) The rate of heat rejection to the cooling water and its temperature rise are

$$\dot{Q}_{\text{out}} = \dot{m}q_{\text{out}} = (41.43 \text{ kg/s})(2125.3 \text{ kJ/kg}) = 88,051 \text{ kJ/s}$$

$$\Delta T_{\text{cooling water}} = \frac{\dot{Q}_{\text{out}}}{(\dot{m}c)_{\text{cooling water}}} = \frac{88,051 \text{ kJ/s}}{(2000 \text{ kg/s})(4.18 \text{ kJ/kg} \cdot ^\circ\text{C})} = \mathbf{10.5^\circ\text{C}}$$

10-24 The net work output and the thermal efficiency for the Carnot and the simple ideal Rankine cycles with steam as the working fluid are to be calculated and compared. Steam enters the turbine in both cases at 10 MPa as a saturated vapor, and the condenser pressure is 20 kPa. In the Rankine cycle, the condenser exit state is saturated liquid and in the Carnot cycle, the boiler inlet state is saturated liquid. Draw the T - s diagrams for both cycles.

10–26 The schematic of a single-flash geothermal power plant with state numbers is given in Fig. P10–26. Geothermal resource exists as saturated liquid at 230°C. The geothermal liquid is withdrawn from the production well at a rate of 230 kg/s, and is flashed to a pressure of 500 kPa by an essentially isenthalpic flashing process where the resulting vapor is separated from the liquid in a separator and directed to the turbine. The steam leaves the turbine at 10 kPa with a moisture content of 10 percent and enters the condenser where it is condensed and routed to a reinjection well along with the liquid coming off the separator. Determine (a) the mass flow rate of steam through the turbine, (b) the isentropic efficiency of the turbine, (c) the power output of the turbine, and (d) the thermal efficiency of the plant (the ratio of the turbine work output to the energy of the geothermal fluid relative to standard ambient conditions). *Answers: (a) 38.2 kg/s, (b) 0.686, (c) 15.4 MW, (d) 7.6 percent*

Analysis (a) We use properties of water for geothermal water (Tables A-4 through A-6)

$$\left. \begin{array}{l} T_1 = 230^\circ\text{C} \\ x_1 = 0 \end{array} \right\} h_1 = 990.14 \text{ kJ/kg}$$

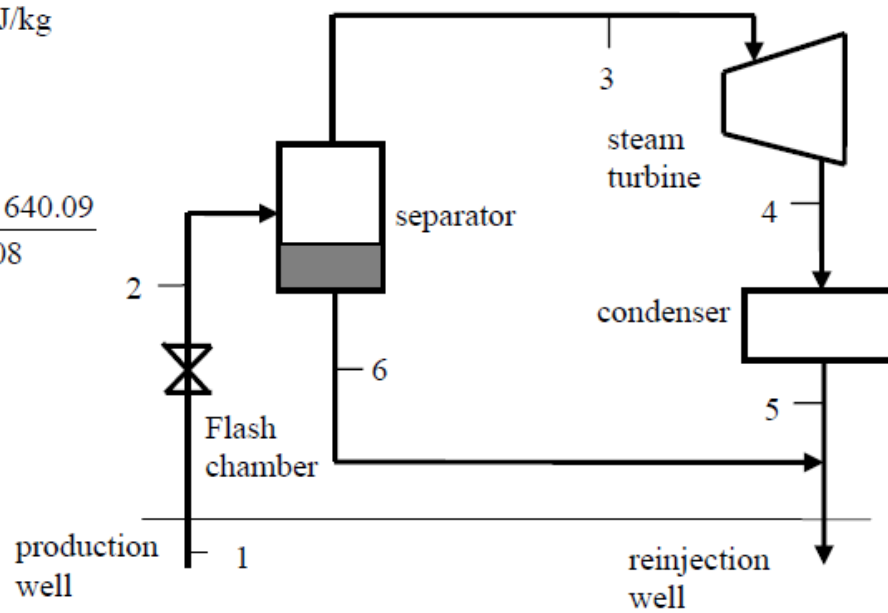
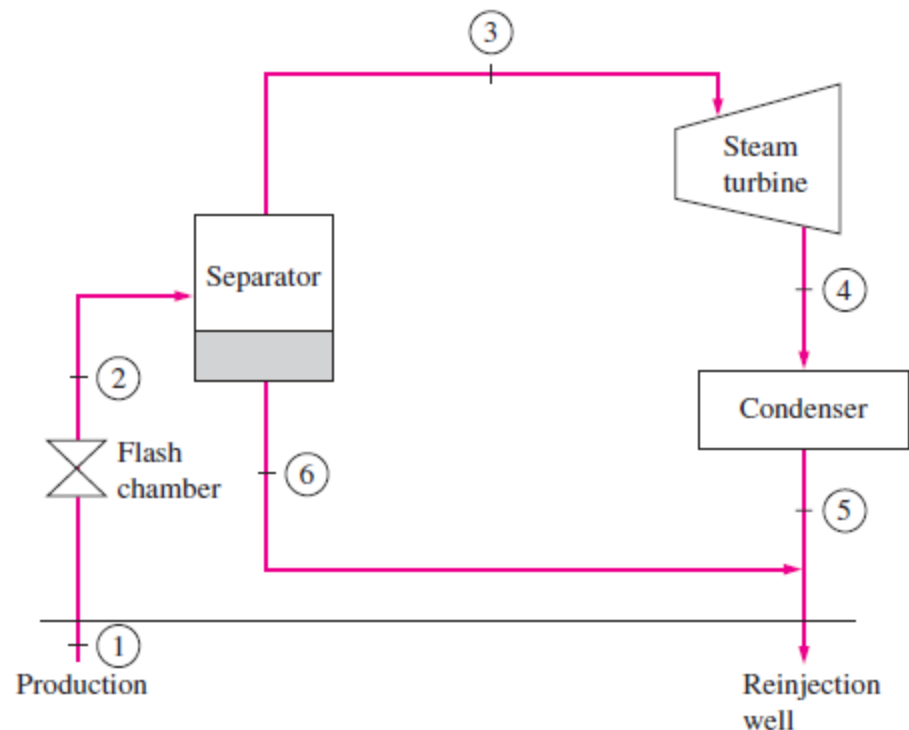
$$\left. \begin{array}{l} P_2 = 500 \text{ kPa} \\ h_2 = h_1 = 990.14 \text{ kJ/kg} \end{array} \right\} x_2 = \frac{h_2 - h_f}{h_{fg}}$$

$$= \frac{990.14 - 640.09}{2108}$$

$$= 0.1661$$

The mass flow rate of steam through the turbine is

$$\begin{aligned} \dot{m}_3 &= x_2 \dot{m}_1 \\ &= (0.1661)(230 \text{ kg/s}) \\ &= \mathbf{38.20 \text{ kg/s}} \end{aligned}$$



(b) Turbine:

$$\left. \begin{array}{l} P_3 = 500 \text{ kPa} \\ x_3 = 1 \end{array} \right\} \begin{array}{l} h_3 = 2748.1 \text{ kJ/kg} \\ s_3 = 6.8207 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_4 = 10 \text{ kPa} \\ s_4 = s_3 \end{array} \right\} h_{4s} = 2160.3 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_4 = 10 \text{ kPa} \\ x_4 = 0.90 \end{array} \right\} h_4 = h_f + x_4 h_{fg} = 191.81 + (0.90)(2392.1) = 2344.7 \text{ kJ/kg}$$

$$\eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}} = \frac{2748.1 - 2344.7}{2748.1 - 2160.3} = \mathbf{0.686}$$

(c) The power output from the turbine is

$$\dot{W}_{T,\text{out}} = \dot{m}_3 (h_3 - h_4) = (38.20 \text{ kJ/kg})(2748.1 - 2344.7) \text{ kJ/kg} = \mathbf{15,410 \text{ kW}}$$

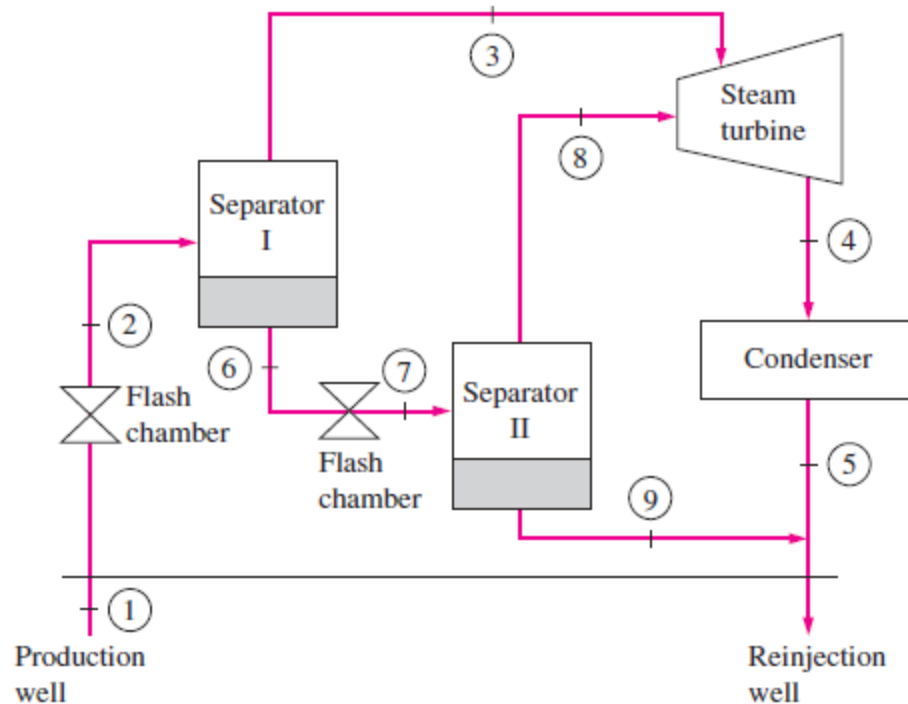
(d) We use saturated liquid state at the standard temperature for dead state enthalpy

$$\left. \begin{array}{l} T_0 = 25^\circ\text{C} \\ x_0 = 0 \end{array} \right\} h_0 = 104.83 \text{ kJ/kg}$$

$$\dot{E}_{\text{in}} = \dot{m}_1 (h_1 - h_0) = (230 \text{ kJ/kg})(990.14 - 104.83) \text{ kJ/kg} = 203,622 \text{ kW}$$

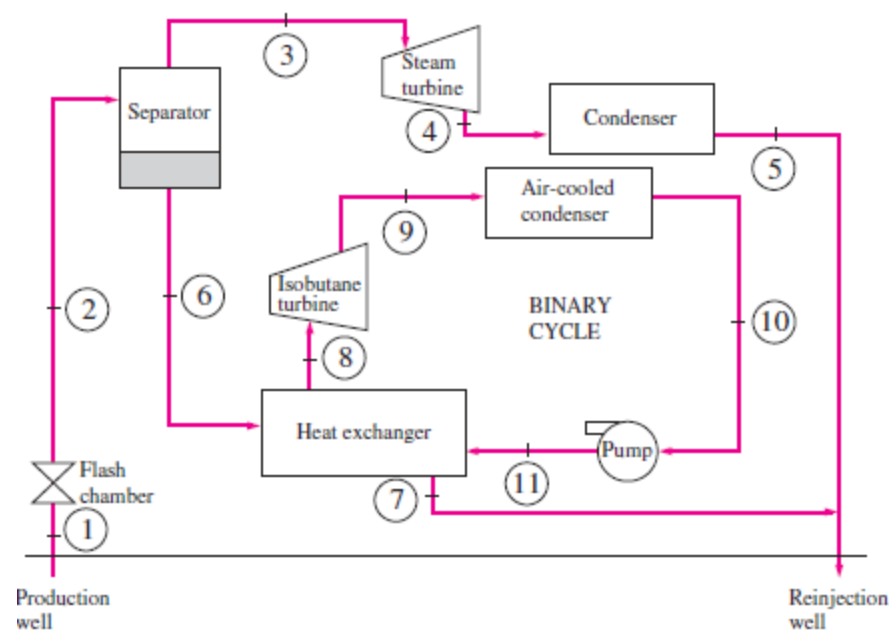
$$\eta_{\text{th}} = \frac{\dot{W}_{T,\text{out}}}{\dot{E}_{\text{in}}} = \frac{15,410}{203,622} = 0.0757 = \mathbf{7.6\%}$$

10-27 Reconsider Prob. 10-26. Now, it is proposed that the liquid water coming out of the separator be routed through another flash chamber maintained at 150 kPa, and the steam produced be directed to a lower stage of the same turbine. Both streams of steam leave the turbine at the same state of 10 kPa and 90 percent quality. Determine (a) the temperature of steam at the outlet of the second flash chamber, (b) the power produced by the lower stage of the turbine, and (c) the thermal efficiency of the plant.



10-28 Reconsider Prob. 10-26. Now, it is proposed that the liquid water coming out of the separator be used as the heat source in a binary cycle with isobutane as the working fluid. Geothermal liquid water leaves the heat exchanger at 90°C while isobutane enters the turbine at 3.25 MPa and 145°C and leaves at 80°C and 400 kPa. Isobutane is condensed in an air-cooled condenser and then pumped to the heat exchanger pressure. Assuming an isentropic efficiency of 90 percent for the pump, determine (a) the mass flow rate of isobutane in the binary cycle, (b) the net power outputs of both the flashing and the binary sections of the plant, and (c) the thermal efficiencies of the binary cycle and the combined plant.

Answers: (a) 105.5 kg/s, (b) 15.4 MW, 6.14 MW, (c) 12.2 percent, 10.6 percent



The Reheat Rankine Cycle

10-32



A steam power plant operates on the ideal reheat Rankine cycle. Steam enters the high-pressure turbine at 8 MPa and 500°C and leaves at 3 MPa. Steam is then reheated at constant pressure to 500°C before it expands to 20 kPa in the low-pressure turbine. Determine the turbine work output, in kJ/kg, and the thermal efficiency of the cycle. Also, show the cycle on a T - s diagram with respect to saturation lines.

Analysis From the steam tables (Tables A-4, A-5, and A-6),

$$h_1 = h_f @ 20 \text{ kPa} = 251.42 \text{ kJ/kg}$$

$$v_1 = v_f @ 20 \text{ kPa} = 0.001017 \text{ m}^3/\text{kg}$$

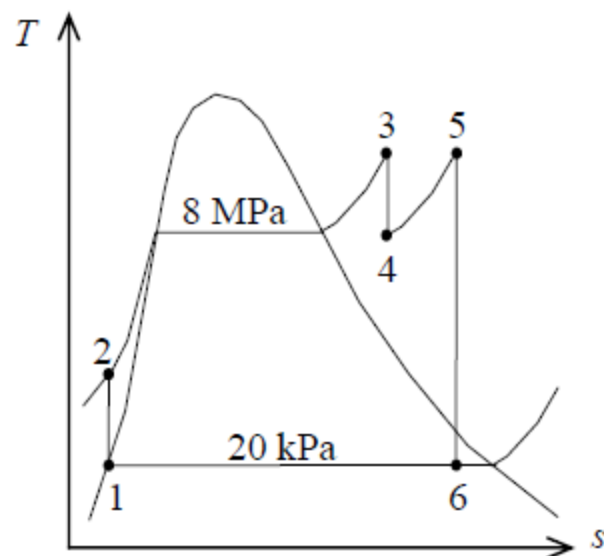
$$\begin{aligned} w_{p,\text{in}} &= v_1(P_2 - P_1) \\ &= (0.001017 \text{ m}^3/\text{kg})(8000 - 20 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 8.12 \text{ kJ/kg} \end{aligned}$$

$$h_2 = h_1 + w_{p,\text{in}} = 251.42 + 8.12 = 259.54 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_3 = 8 \text{ MPa} \\ T_3 = 500^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 = 3399.5 \text{ kJ/kg} \\ s_3 = 6.7266 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_4 = 3 \text{ MPa} \\ s_4 = s_3 \end{array} \right\} h_4 = 3105.1 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_5 = 3 \text{ MPa} \\ T_5 = 500^\circ\text{C} \end{array} \right\} \begin{array}{l} h_5 = 3457.2 \text{ kJ/kg} \\ s_5 = 7.2359 \text{ kJ/kg} \cdot \text{K} \end{array}$$



$$\left. \begin{array}{l} P_6 = 20 \text{ kPa} \\ s_6 = s_5 \end{array} \right\} \begin{array}{l} x_6 = \frac{s_6 - s_f}{s_{fg}} = \frac{7.2359 - 0.8320}{7.0752} = 0.9051 \\ h_6 = h_f + x_6 h_{fg} = 251.42 + (0.9051)(2357.5) = 2385.2 \text{ kJ/kg} \end{array}$$

The turbine work output and the thermal efficiency are determined from

$$w_{T,\text{out}} = (h_3 - h_4) + (h_5 - h_6) = 3399.5 - 3105.1 + 3457.2 - 2385.2 = \mathbf{1366.4 \text{ kJ/kg}}$$

and

$$q_{\text{in}} = (h_3 - h_2) + (h_5 - h_4) = 3399.5 - 259.54 + 3457.2 - 3105.1 = 3492.0 \text{ kJ/kg}$$

$$w_{\text{net}} = w_{T,\text{out}} - w_{p,\text{in}} = 1366.4 - 8.12 = 1358.3 \text{ kJ/kg}$$

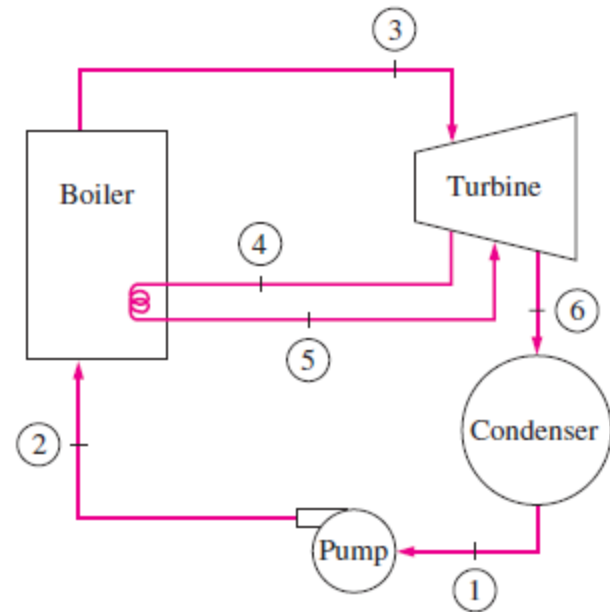
Thus,

$$\eta_{\text{th}} = \frac{w_{\text{net}}}{q_{\text{in}}} = \frac{1358.3 \text{ kJ/kg}}{3492.5 \text{ kJ/kg}} = \mathbf{38.9\%}$$

10–37 A steam power plant operates on an ideal reheat Rankine cycle between the pressure limits of 15 MPa and 10 kPa. The mass flow rate of steam through the cycle is 12 kg/s. Steam enters both stages of the turbine at 500°C. If the moisture content of the steam at the exit of the low-pressure turbine is not to exceed 10 percent, determine (a) the pressure at which reheating takes place, (b) the total rate of heat input in the boiler, and (c) the thermal efficiency of the cycle. Also, show the cycle on a T - s diagram with respect to saturation lines.

10-38 A steam power plant operates on the reheat Rankine cycle. Steam enters the high-pressure turbine at 12.5 MPa and 550°C at a rate of 7.7 kg/s and leaves at 2 MPa. Steam is then reheated at constant pressure to 450°C before it expands in the low-pressure turbine. The isentropic efficiencies of the turbine and the pump are 85 percent and 90 percent, respectively. Steam leaves the condenser as a saturated liquid. If the moisture content of the steam at the exit of the turbine is not to exceed 5 percent, determine (a) the condenser pressure, (b) the net power output, and (c) the thermal efficiency.

Answers: (a) 9.73 kPa, (b) 10.2 MW, (c) 36.9 percent



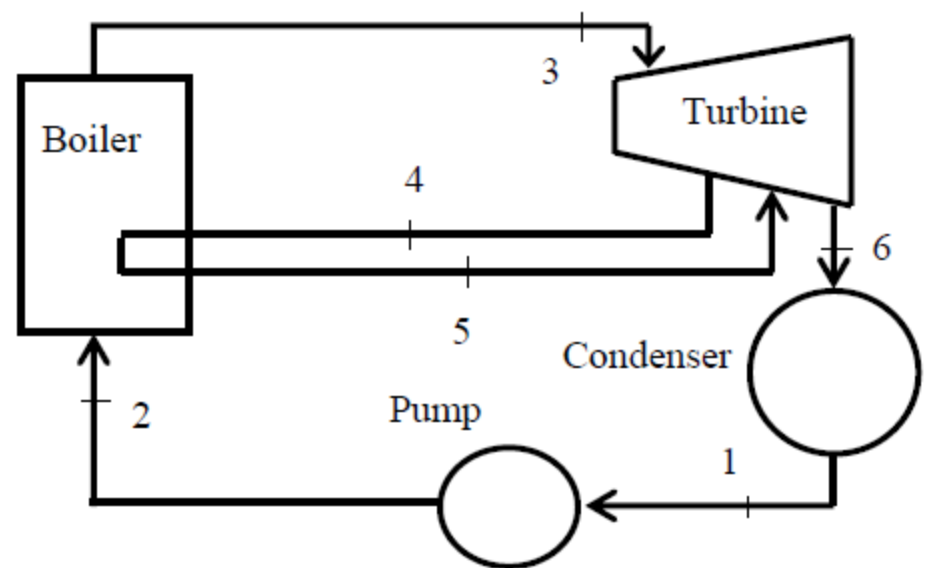
Analysis (a) From the steam tables (Tables A-4, A-5, and A-6),

$$\left. \begin{array}{l} P_3 = 12.5 \text{ MPa} \\ T_3 = 550^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 = 3476.5 \text{ kJ/kg} \\ s_3 = 6.6317 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_4 = 2 \text{ MPa} \\ s_{4s} = s_3 \end{array} \right\} h_{4s} = 2948.1 \text{ kJ/kg}$$

$$\eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}}$$

$$\begin{aligned} \rightarrow h_4 &= h_3 - \eta_T (h_3 - h_{4s}) \\ &= 3476.5 - (0.85)(3476.5 - 2948.1) \\ &= 3027.3 \text{ kJ/kg} \end{aligned}$$

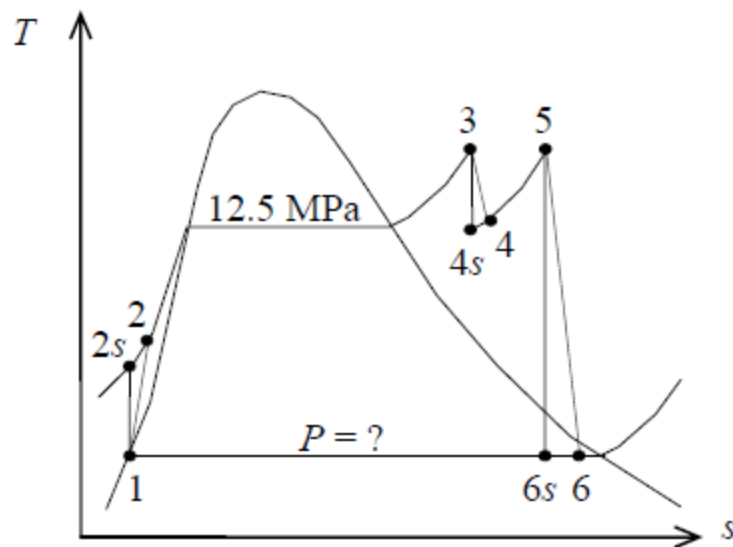


$$\left. \begin{array}{l} P_5 = 2 \text{ MPa} \\ T_5 = 450^\circ\text{C} \end{array} \right\} \begin{array}{l} h_5 = 3358.2 \text{ kJ/kg} \\ s_5 = 7.2815 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_6 = ? \\ x_6 = 0.95 \end{array} \right\} h_6 =$$

$$\left. \begin{array}{l} P_6 = ? \\ s_6 = s_5 \end{array} \right\} h_{6s} =$$

$$\begin{aligned} \eta_T &= \frac{h_5 - h_6}{h_5 - h_{6s}} \longrightarrow h_6 = h_5 - \eta_T (h_5 - h_{6s}) \\ &= 3358.2 - (0.85)(3358.2 - 2948.1) \\ &= 3027.3 \text{ kJ/kg} \end{aligned}$$



The pressure at state 6 may be determined by a trial-error approach from the steam tables or by using EES from the above equations:

$$P_6 = \mathbf{9.73 \text{ kPa}}, \quad h_6 = 2463.3 \text{ kJ/kg},$$

(b) Then,

$$h_1 = h_f @ 9.73 \text{ kPa} = 189.57 \text{ kJ/kg}$$

$$v_1 = v_f @ 10 \text{ kPa} = 0.001010 \text{ m}^3/\text{kg}$$

$$\begin{aligned} w_{p,\text{in}} &= v_1 (P_2 - P_1) / \eta_p \\ &= (0.00101 \text{ m}^3/\text{kg}) (12,500 - 9.73 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) / (0.90) \\ &= 14.02 \text{ kJ/kg} \end{aligned}$$

$$h_2 = h_1 + w_{p,\text{in}} = 189.57 + 14.02 = 203.59 \text{ kJ/kg}$$



Cycle analysis:

$$q_{\text{in}} = (h_3 - h_2) + (h_5 - h_4) = 3476.5 - 3027.3 + 3358.2 - 2463.3 = 3603.8 \text{ kJ/kg}$$

$$q_{\text{out}} = h_6 - h_1 = 3027.3 - 189.57 = 2273.7 \text{ kJ/kg}$$

$$\dot{W}_{\text{net}} = \dot{m}(q_{\text{in}} - q_{\text{out}}) = (7.7 \text{ kg/s})(3603.8 - 2273.7) \text{ kJ/kg} = \mathbf{10,242 \text{ kW}}$$

(c) The thermal efficiency is

$$\eta_{\text{th}} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{2273.7 \text{ kJ/kg}}{3603.8 \text{ kJ/kg}} = 0.369 = \mathbf{36.9\%}$$

10–46 A steam power plant operates on an ideal regenerative Rankine cycle with two open feedwater heaters. Steam enters the turbine at 10 MPa and 600°C and exhausts to the condenser at 5 kPa. Steam is extracted from the turbine at 0.6 and 0.2 MPa. Water leaves both feedwater heaters as a saturated liquid. The mass flow rate of steam through the boiler is 22 kg/s. Show the cycle on a T - s diagram, and determine (a) the net power output of the power plant and (b) the thermal efficiency of the cycle. *Answers: (a) 30.5 MW, (b) 47.1 percent*

10-47



10-47  Consider an ideal steam regenerative Rankine cycle with two feedwater heaters, one closed and one open. Steam enters the turbine at 12.5 MPa and 550°C and exhausts to the condenser at 10 kPa. Steam is extracted from the turbine at 0.8 MPa for the closed feedwater heater and at 0.3 MPa for the open one. The feedwater is heated to the condensation temperature of the extracted steam in the closed feedwater heater. The extracted steam leaves the closed feedwater heater as a saturated liquid, which is subsequently throttled to the open feedwater heater. Show the cycle on a T - s diagram with respect to saturation lines, and determine (a) the mass flow rate of steam through the boiler for a net power output of 250 MW and (b) the thermal efficiency of the cycle.

Analysis (a) From the steam tables (Tables A-4, A-5, and A-6),

$$h_1 = h_{f@10 \text{ kPa}} = 191.81 \text{ kJ/kg}$$

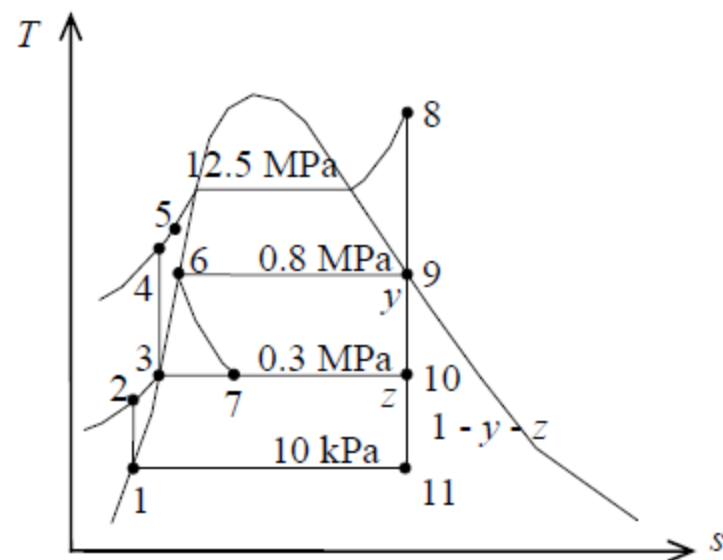
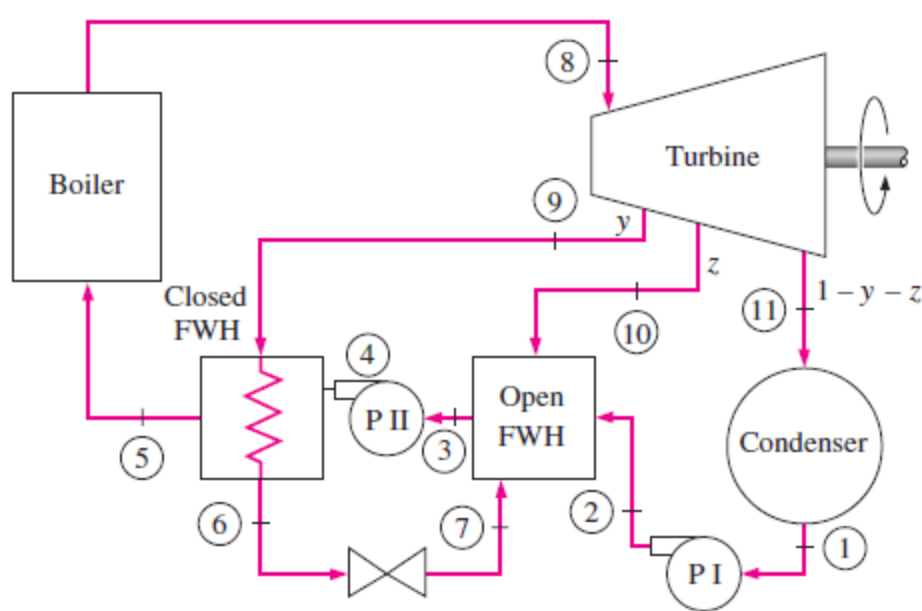
$$\nu_1 = \nu_{f@10 \text{ kPa}} = 0.00101 \text{ m}^3/\text{kg}$$

$$\begin{aligned} w_{PI,in} &= v_1(P_2 - P_1) \\ &= (0.00101 \text{ m}^3/\text{kg})(300 - 10 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 0.29 \text{ kJ/kg} \end{aligned}$$

$$h_2 = h_1 + w_{pI,\text{in}} = 191.81 + 0.29 = 192.10 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_3 = 0.3 \text{ MPa} \\ \text{sat. liquid} \end{array} \right\} \begin{array}{l} h_3 = h_{f@0.3 \text{ MPa}} = 561.43 \text{ kJ/kg} \\ \nu_3 = \nu_{f@0.3 \text{ MPa}} = 0.001073 \text{ m}^3/\text{kg} \end{array}$$

$$\begin{aligned} w_{pII, \text{in}} &= v_3(P_4 - P_3) \\ &= (0.001073 \text{ m}^3/\text{kg})(12,500 - 300 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 13.09 \text{ kJ/kg} \end{aligned}$$



$$h_4 = h_3 + w_{pII,in} = 561.43 + 13.09 = 574.52 \text{ kJ/kg}$$

$$P_6 = 0.8 \text{ MPa} \left\{ \begin{array}{l} h_6 = h_7 = h_f @ 0.8 \text{ MPa} = 720.87 \text{ kJ/kg} \\ \nu_6 = \nu_f @ 0.8 \text{ MPa} = 0.001115 \text{ m}^3/\text{kg} \\ T_6 = T_{\text{sat}} @ 0.8 \text{ MPa} = 170.4^\circ\text{C} \end{array} \right.$$

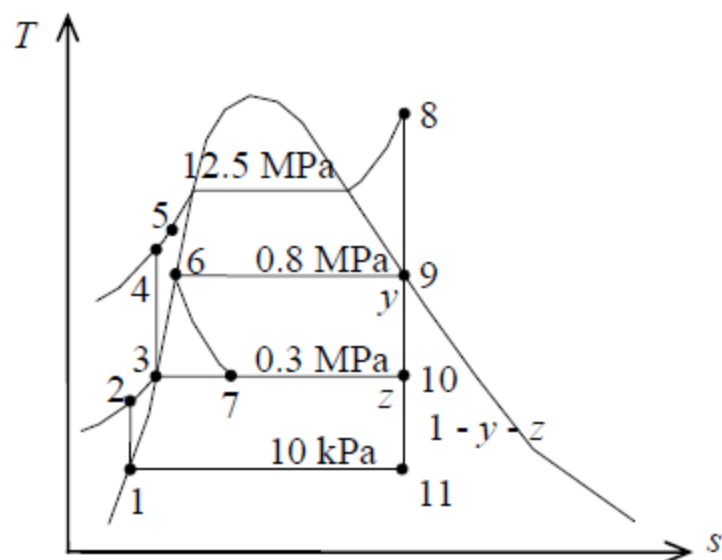
$$T_6 = T_5, P_5 = 12.5 \text{ MPa} \rightarrow h_5 = 727.83 \text{ kJ/kg}$$

$$P_8 = 12.5 \text{ MPa} \left\{ \begin{array}{l} h_8 = 3476.5 \text{ kJ/kg} \\ T_8 = 550^\circ\text{C} \end{array} \right. \left\{ \begin{array}{l} s_8 = 6.6317 \text{ kJ/kg} \cdot \text{K} \end{array} \right.$$

$$P_9 = 0.8 \text{ MPa} \left\{ \begin{array}{l} x_9 = \frac{s_9 - s_f}{s_{fg}} = \frac{6.6317 - 2.0457}{4.6160} = 0.9935 \\ s_9 = s_8 \end{array} \right. \left\{ \begin{array}{l} h_9 = h_f + x_9 h_{fg} = 720.87 + (0.9935)(2047.5) = 2755.0 \text{ kJ/kg} \end{array} \right.$$

$$P_{10} = 0.3 \text{ MPa} \left\{ \begin{array}{l} x_{10} = \frac{s_{10} - s_f}{s_{fg}} = \frac{6.6317 - 1.6717}{5.3200} = 0.9323 \\ s_{10} = s_8 \end{array} \right. \left\{ \begin{array}{l} h_{10} = h_f + x_{10} h_{fg} = 561.43 + (0.9323)(2163.5) = 2578.5 \text{ kJ/kg} \end{array} \right.$$

$$P_{11} = 10 \text{ kPa} \left\{ \begin{array}{l} x_{11} = \frac{s_{11} - s_f}{s_{fg}} = \frac{6.6317 - 0.6492}{7.4996} = 0.7977 \\ s_{11} = s_8 \end{array} \right. \left\{ \begin{array}{l} h_{11} = h_f + x_{11} h_{fg} = 191.81 + (0.7977)(2392.1) = 2100.0 \text{ kJ/kg} \end{array} \right.$$



The fraction of steam extracted is determined from the steady-flow energy balance equation applied to the feedwater heaters. Noting that $\dot{Q} \cong \dot{W} \cong \Delta \text{ke} \cong \Delta \text{pe} \cong 0$,

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \overset{\text{0 (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e \longrightarrow \dot{m}_9 (h_9 - h_6) = \dot{m}_5 (h_5 - h_4) \longrightarrow y(h_9 - h_6) = (h_5 - h_4)$$

where y is the fraction of steam extracted from the turbine ($= \dot{m}_{10} / \dot{m}_5$). Solving for y ,

$$y = \frac{h_5 - h_4}{h_9 - h_6} = \frac{727.83 - 574.52}{2755.0 - 720.87} = 0.0753$$

For the open FWH,

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \overset{\text{0 (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e \longrightarrow \dot{m}_7 h_7 + \dot{m}_2 h_2 + \dot{m}_{10} h_{10} = \dot{m}_3 h_3 \longrightarrow y h_7 + (1 - y - z) h_2 + z h_{10} = (1) h_3$$

where z is the fraction of steam extracted from the turbine ($= \dot{m}_9 / \dot{m}_5$) at the second stage. Solving for z ,

$$z = \frac{(h_3 - h_2) - y(h_7 - h_2)}{h_{10} - h_2} = \frac{561.43 - 192.10 - (0.0753)(720.87 - 192.10)}{2578.5 - 192.10} = 0.1381$$

Then,

$$q_{\text{in}} = h_8 - h_5 = 3476.5 - 727.36 = 2749.1 \text{ kJ/kg}$$

$$q_{\text{out}} = (1 - y - z)(h_{11} - h_1) = (1 - 0.0753 - 0.1381)(2100.0 - 191.81) = 1500.1 \text{ kJ/kg}$$

$$w_{\text{net}} = q_{\text{in}} - q_{\text{out}} = 2749.1 - 1500.1 = 1249 \text{ kJ/kg}$$

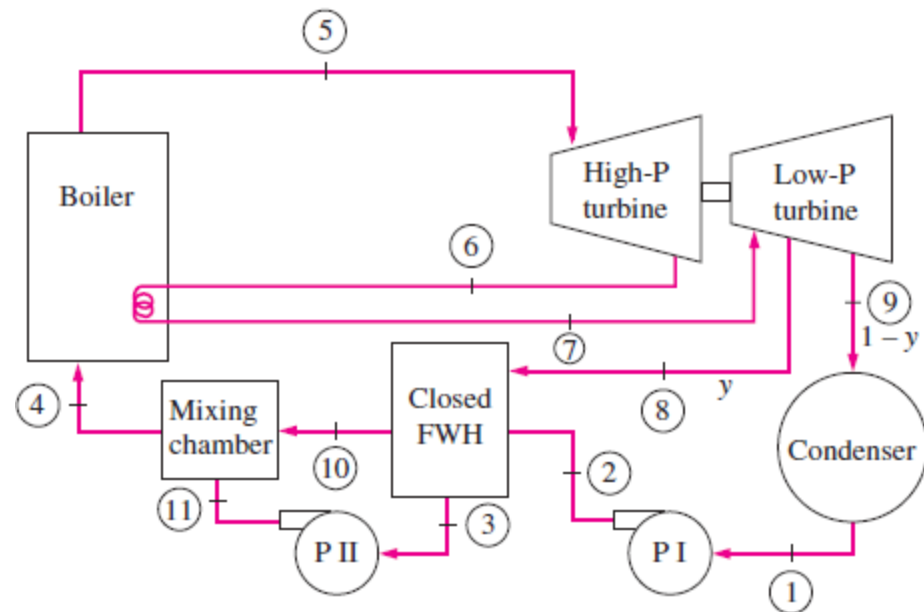


and

$$\dot{m} = \frac{\dot{W}_{\text{net}}}{w_{\text{net}}} = \frac{250,000 \text{ kJ/s}}{1249 \text{ kJ/kg}} = \mathbf{200.2 \text{ kg/s}}$$

$$(b) \quad \eta_{\text{th}} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{1500.1 \text{ kJ/kg}}{2749.1 \text{ kJ/kg}} = \mathbf{45.4\%}$$

10-52 A steam power plant operates on the reheat-regenerative Rankine cycle with a closed feedwater heater. Steam enters the turbine at 12.5 MPa and 550°C at a rate of 24 kg/s and is condensed in the condenser at a pressure of 20 kPa. Steam is reheated at 5 MPa to 550°C. Some steam is extracted from the low-pressure turbine at 1.0 MPa, is completely condensed in the closed feedwater heater, and pumped to 12.5 MPa before it mixes with the feedwater at the same pressure. Assuming an isentropic efficiency of 88 percent for both the turbine and the pump, determine (a) the temperature of the steam at the inlet of the closed feedwater heater, (b) the mass flow rate of the steam extracted from the turbine for the closed feedwater heater, (c) the net power output, and (d) the thermal efficiency. *Answers: (a) 328°C, (b) 4.29 kg/s, (c) 28.6 MW, (d) 39.3 percent*



10-57 Determine the exergy destruction associated with each of the processes of the reheat Rankine cycle described in Prob. 10-32. Assume a source temperature of 1800 K and a sink temperature of 300 K.

Assumptions 1 Steady operating conditions exist. 2 Kinetic and potential energy changes are negligible.

Analysis From Problem 10-32

$$s_1 = s_2 = s_f @$$

$$s_3 = s_4 =$$

$$s_5 = s_6 =$$

$$q_{\text{in},2-3} = h_3 - h_2 =$$

$$q_{\text{in},4-5} = h_5 - h_4 =$$

$$q_{\text{out}} =$$

The exergy destruction during a process of a stream from an inlet state to exit state is given by

$$x_{\text{dest}} = T_0 s_{\text{gen}} = T_0 \left(s_e - s_i - \frac{q_{\text{in}}}{T_{\text{source}}} + \frac{q_{\text{out}}}{T_{\text{sink}}} \right)$$

Application of this equation for each process of the cycle gives

$$x_{\text{destroyed},23} = T_0 \left(s_3 - s_2 - \frac{q_{\text{in},2-3}}{T_{\text{source}}} \right) =$$

$$x_{\text{destroyed},45} = T_0 \left(s_5 - s_4 - \frac{q_{\text{in},4-5}}{T_{\text{source}}} \right) =$$

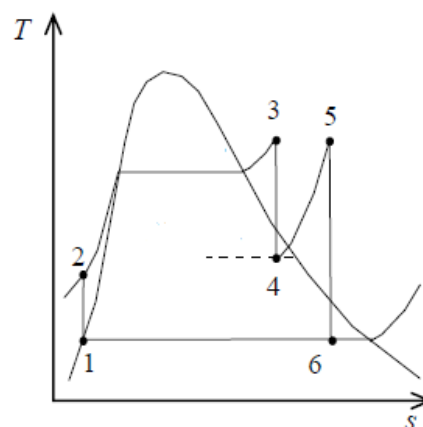
$$x_{\text{destroyed},61} = T_0 \left(s_1 - s_6 + \frac{q_{\text{out}}}{T_{\text{sink}}} \right) =$$

Processes 1-2, 3-4, and 5-6 are isentropic, and thus,

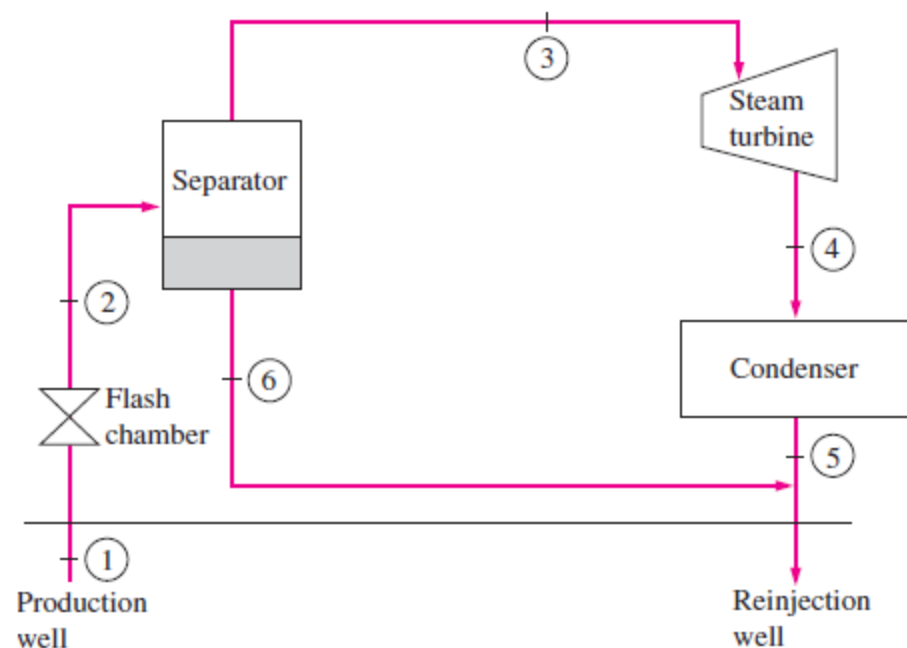
$$x_{\text{destroyed},12} = 0$$

$$x_{\text{destroyed},34} = 0$$

$$x_{\text{destroyed},56} = 0$$



10–62 The schematic of a single-flash geothermal power plant with state numbers is given in Fig. P10–62. Geothermal resource exists as saturated liquid at 230°C . The geothermal liquid is withdrawn from the production well at a rate of 230 kg/s and is flashed to a pressure of 500 kPa by an essentially isenthalpic flashing process where the resulting vapor is separated from the liquid in a separator and is directed to the turbine. The steam leaves the turbine at 10 kPa with a moisture content of 5 percent and enters the condenser where it is condensed; it is routed to a reinjection well along with the liquid coming off the separator. Determine (a) the power output of the turbine and the thermal efficiency of the plant, (b) the exergy of the geothermal liquid at the exit of the flash chamber, and the exergy destructions and the second-law (exergetic) efficiencies for (c) the flash chamber, (d) the turbine, and (e) the entire plant. *Answers: (a) 10.8 MW, 0.053, (b) 17.3 MW, (c) 5.1 MW, 0.898, (d) 10.9 MW, 0.500, (e) 39.0 MW, 0.218*



Analysis (a) We use properties of water for geothermal water (Tables A-4, A-5, and A-6)

$$\left. \begin{array}{l} T_1 = 230^\circ\text{C} \\ x_1 = 0 \end{array} \right\} \begin{array}{l} h_1 = 990.14 \text{ kJ/kg} \\ s_1 = 2.6100 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 500 \text{ kPa} \\ h_2 = h_1 = 990.14 \text{ kJ/kg} \end{array} \right\} \begin{array}{l} x_2 = 0.1661 \\ s_2 = 2.6841 \text{ kJ/kg}\cdot\text{K} \end{array}$$

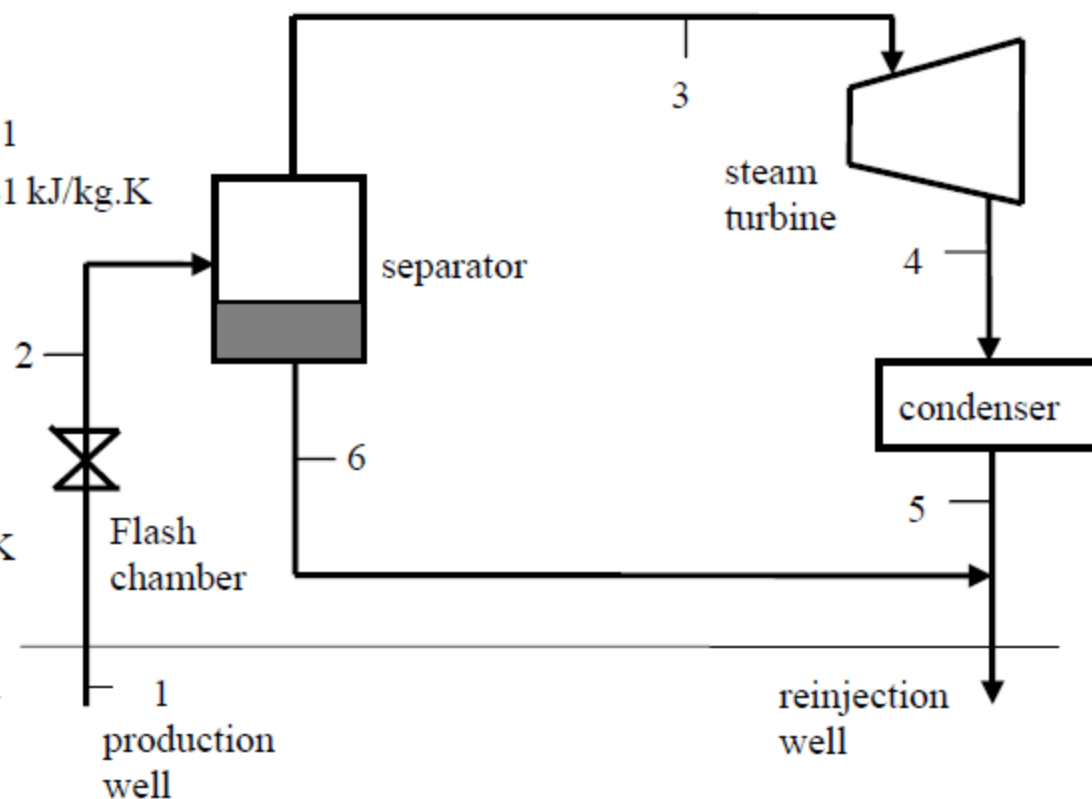
$$\begin{aligned} \dot{m}_3 &= x_2 \dot{m}_1 \\ &= (0.1661)(230 \text{ kg/s}) \\ &= 38.19 \text{ kg/s} \end{aligned}$$

$$\left. \begin{array}{l} P_3 = 500 \text{ kPa} \\ x_3 = 1 \end{array} \right\} \begin{array}{l} h_3 = 2748.1 \text{ kJ/kg} \\ s_3 = 6.8207 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\left. \begin{array}{l} P_4 = 10 \text{ kPa} \\ x_4 = 0.95 \end{array} \right\} \begin{array}{l} h_4 = 2464.3 \text{ kJ/kg} \\ s_4 = 7.7739 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\left. \begin{array}{l} P_6 = 500 \text{ kPa} \\ x_6 = 0 \end{array} \right\} \begin{array}{l} h_6 = 640.09 \text{ kJ/kg} \\ s_6 = 1.8604 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\dot{m}_6 = \dot{m}_1 - \dot{m}_3 = 230 - 38.19 = 191.81 \text{ kg/s}$$



The power output from the turbine is

$$\dot{W}_T = \dot{m}_3 (h_3 - h_4) = (38.19 \text{ kJ/kg})(2748.1 - 2464.3) \text{ kJ/kg} = \mathbf{10,842 \text{ kW}}$$

We use saturated liquid state at the standard temperature for dead state properties

$$\left. \begin{array}{l} T_0 = 25^\circ\text{C} \\ x_0 = 0 \end{array} \right\} \begin{array}{l} h_0 = 104.83 \text{ kJ/kg} \\ s_0 = 0.3672 \text{ kJ/kg} \end{array}$$

$$\dot{E}_{\text{in}} = \dot{m}_1(h_1 - h_0) = (230 \text{ kJ/kg})(990.14 - 104.83) \text{ kJ/kg} = 203,622 \text{ kW}$$

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{T,out}}}{\dot{E}_{\text{in}}} = \frac{10,842}{203,622} = 0.0532 = \mathbf{5.3\%}$$

(b) The specific exergies at various states are

$$\psi_1 = h_1 - h_0 - T_0(s_1 - s_0) = (990.14 - 104.83) \text{ kJ/kg} - (298 \text{ K})(2.6100 - 0.3672) \text{ kJ/kg.K} = 216.53 \text{ kJ/kg}$$

$$\psi_2 = h_2 - h_0 - T_0(s_2 - s_0) = (990.14 - 104.83) \text{ kJ/kg} - (298 \text{ K})(2.6841 - 0.3672) \text{ kJ/kg.K} = 194.44 \text{ kJ/kg}$$

$$\psi_3 = h_3 - h_0 - T_0(s_3 - s_0) = (2748.1 - 104.83) \text{ kJ/kg} - (298 \text{ K})(6.8207 - 0.3672) \text{ kJ/kg.K} = 719.10 \text{ kJ/kg}$$

$$\psi_4 = h_4 - h_0 - T_0(s_4 - s_0) = (2464.3 - 104.83) \text{ kJ/kg} - (298 \text{ K})(7.7739 - 0.3672) \text{ kJ/kg.K} = 151.05 \text{ kJ/kg}$$

$$\psi_6 = h_6 - h_0 - T_0(s_6 - s_0) = (640.09 - 104.83) \text{ kJ/kg} - (298 \text{ K})(1.8604 - 0.3672) \text{ kJ/kg.K} = 89.97 \text{ kJ/kg}$$



The exergy of geothermal water at state 6 is

$$\dot{X}_6 = \dot{m}_6 \psi_6 = (191.81 \text{ kg/s})(89.97 \text{ kJ/kg}) = \mathbf{17,257 \text{ kW}}$$

(c) Flash chamber:

$$\dot{X}_{\text{dest, FC}} = \dot{m}_1 (\psi_1 - \psi_2) = (230 \text{ kg/s})(216.53 - 194.44) \text{ kJ/kg} = \mathbf{5080 \text{ kW}}$$

$$\eta_{\text{II, FC}} = \frac{\psi_2}{\psi_1} = \frac{194.44}{216.53} = 0.898 = \mathbf{89.8\%}$$

(d) Turbine:

$$\dot{X}_{\text{dest, T}} = \dot{m}_3 (\psi_3 - \psi_4) - \dot{W}_T = (38.19 \text{ kg/s})(719.10 - 151.05) \text{ kJ/kg} - 10,842 \text{ kW} = \mathbf{10,854 \text{ kW}}$$

$$\eta_{\text{II, T}} = \frac{\dot{W}_T}{\dot{m}_3 (\psi_3 - \psi_4)} = \frac{10,842 \text{ kW}}{(38.19 \text{ kg/s})(719.10 - 151.05) \text{ kJ/kg}} = 0.500 = \mathbf{50.0\%}$$

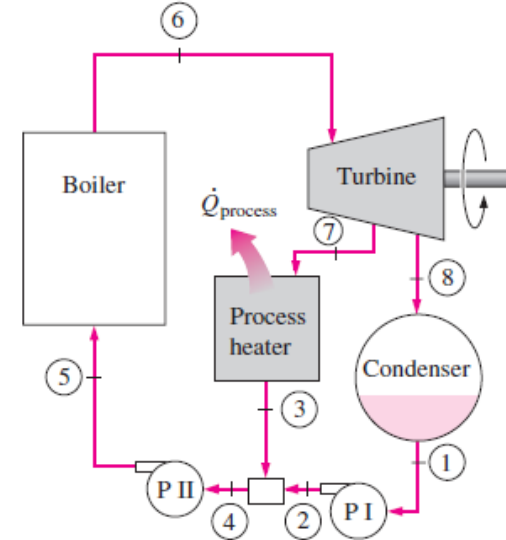
(e) Plant:

$$\dot{X}_{\text{in, Plant}} = \dot{m}_1 \psi_1 = (230 \text{ kg/s})(216.53 \text{ kJ/kg}) = 49,802 \text{ kW}$$

$$\dot{X}_{\text{dest, Plant}} = \dot{X}_{\text{in, Plant}} - \dot{W}_T = 49,802 - 10,842 = \mathbf{38,960 \text{ kW}}$$

$$\eta_{\text{II, Plant}} = \frac{\dot{W}_T}{\dot{X}_{\text{in, Plant}}} = \frac{10,842 \text{ kW}}{49,802 \text{ kW}} = 0.2177 = \mathbf{21.8\%}$$

10–67 Steam enters the turbine of a cogeneration plant at 7 MPa and 500°C. One-fourth of the steam is extracted from the turbine at 600-kPa pressure for process heating. The remaining steam continues to expand to 10 kPa. The extracted steam is then condensed and mixed with feedwater at constant pressure and the mixture is pumped to the boiler pressure of 7 MPa. The mass flow rate of steam through the boiler is 30 kg/s. Disregarding any pressure drops and heat losses in the piping, and assuming the turbine and the pump to be isentropic, determine the net power produced and the utilization factor of the plant.



Analysis From the steam tables (Tables A-4, A-5, and A-6),

$$h_1 = h_f @ 10 \text{ kPa} = 191.81 \text{ kJ/kg}$$

$$\nu_1 = \nu_f @ 10 \text{ kPa} = 0.00101 \text{ m}^3/\text{kg}$$

$$\begin{aligned} w_{pI, \text{in}} &= \nu_1 (P_2 - P_1) \\ &= (0.00101 \text{ m}^3/\text{kg})(600 - 10 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 0.60 \text{ kJ/kg} \end{aligned}$$

$$h_2 = h_1 + w_{pI, \text{in}} = 191.81 + 0.60 = 192.41 \text{ kJ/kg}$$

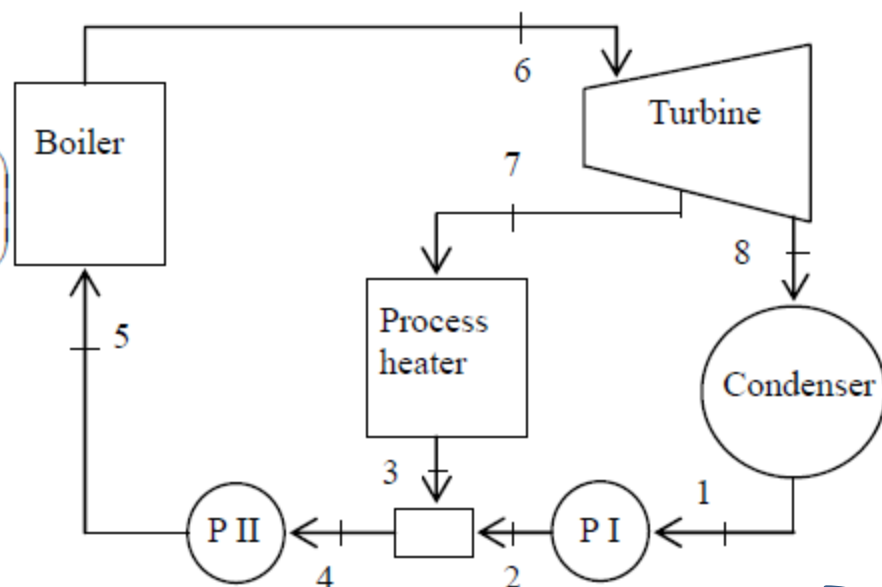
$$h_3 = h_f @ 0.6 \text{ MPa} = 670.38 \text{ kJ/kg}$$

Mixing chamber:

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \xrightarrow{\text{no (steady)}} 0 \longrightarrow \dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e \longrightarrow \dot{m}_4 h_4 = \dot{m}_2 h_2 + \dot{m}_3 h_3$$

$$\text{or, } h_4 = \frac{\dot{m}_2 h_2 + \dot{m}_3 h_3}{\dot{m}_4} = \frac{(22.50)(192.41) + (7.50)(670.38)}{30} = 311.90 \text{ kJ/kg}$$



$$v_4 \cong v_f @ h_f = 311.90 \text{ kJ/kg} = 0.001026 \text{ m}^3/\text{kg}$$

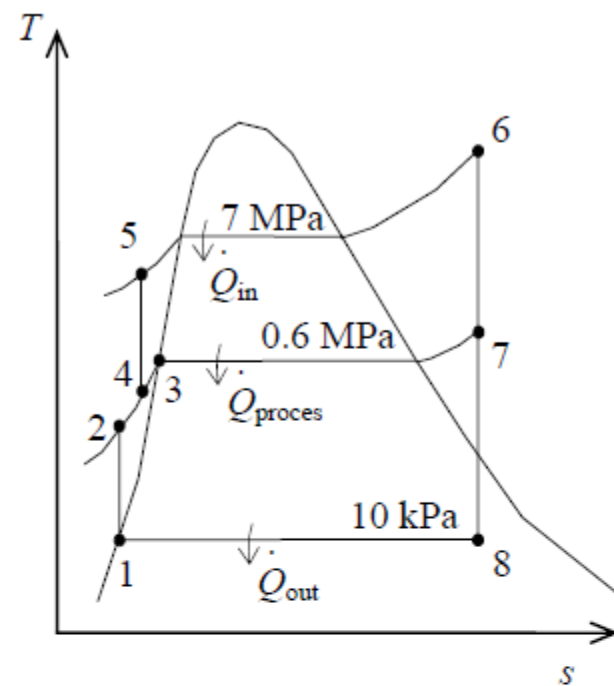
$$\begin{aligned} w_{pII,in} &= v_4(P_5 - P_4) \\ &= (0.001026 \text{ m}^3/\text{kg})(7000 - 600 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 6.57 \text{ kJ/kg} \end{aligned}$$

$$h_5 = h_4 + w_{pII,in} = 311.90 + 6.57 = 318.47 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_6 = 7 \text{ MPa} \\ T_6 = 500^\circ\text{C} \end{array} \right\} \begin{array}{l} h_6 = 3411.4 \text{ kJ/kg} \\ s_6 = 6.8000 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_7 = 0.6 \text{ MPa} \\ s_7 = s_6 \end{array} \right\} h_7 = 2774.6 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_8 = 10 \text{ kPa} \\ s_8 = s_6 \end{array} \right\} \begin{array}{l} x_8 = \frac{s_8 - s_f}{s_{fg}} = \frac{6.8000 - 0.6492}{7.4996} = 0.8201 \\ h_8 = h_f + x_8 h_{fg} = 191.81 + (0.8201)(2392.1) = 2153.6 \text{ kJ/kg} \end{array}$$



Then,

$$\begin{aligned} \dot{W}_{T,out} &= \dot{m}_6(h_6 - h_7) + \dot{m}_8(h_7 - h_8) \\ &= (30 \text{ kg/s})(3411.4 - 2774.6) \text{ kJ/kg} + (22.5 \text{ kg/s})(2774.6 - 2153.6) \text{ kJ/kg} = 33,077 \text{ kW} \end{aligned}$$

$$\dot{W}_{p,in} = \dot{m}_1 w_{pI,in} + \dot{m}_4 w_{pII,in} = (22.5 \text{ kg/s})(0.60 \text{ kJ/kg}) + (30 \text{ kg/s})(6.57 \text{ kJ/kg}) = 210.6 \text{ kW}$$

$$\dot{W}_{net} = \dot{W}_{T,out} - \dot{W}_{p,in} = 33,077 - 210.6 = \mathbf{32,866 \text{ kW}}$$

Also, $\dot{Q}_{process} = \dot{m}_7(h_7 - h_3) = (7.5 \text{ kg/s})(2774.6 - 670.38) \text{ kJ/kg} = 15,782 \text{ kW}$

$$\dot{Q}_{in} = \dot{m}_5(h_6 - h_5) = (30 \text{ kg/s})(3411.4 - 318.47) = 92,788 \text{ kW}$$

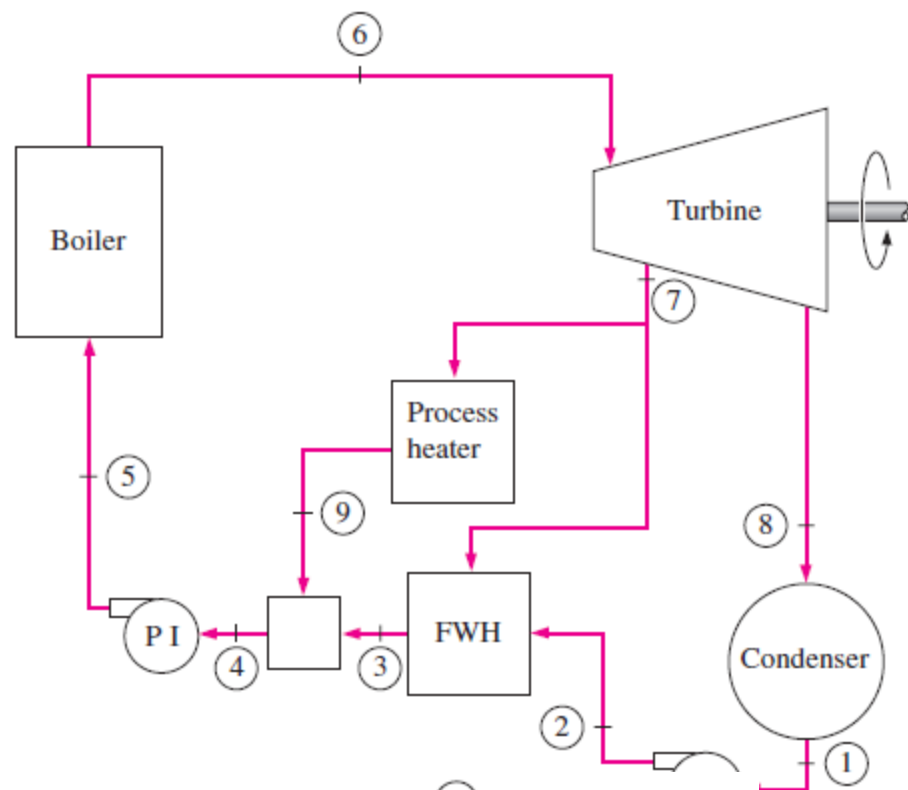
and $\varepsilon_u = \frac{\dot{W}_{net} + \dot{Q}_{process}}{\dot{Q}_{in}} = \frac{32,866 + 15,782}{92,788} = \mathbf{52.4\%}$

10–69 Steam is generated in the boiler of a cogeneration plant at 10 MPa and 450°C at a steady rate of 5 kg/s. In normal operation, steam expands in a turbine to a pressure of 0.5 MPa and is then routed to the process heater, where it supplies the process heat. Steam leaves the process heater as a saturated liquid and is pumped to the boiler pressure. In this mode, no steam passes through the condenser, which operates at 20 kPa.

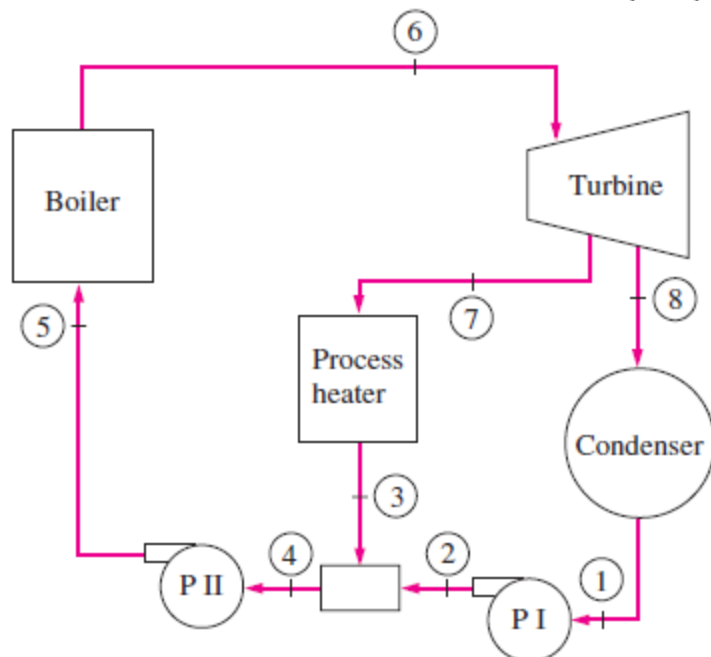
(a) Determine the power produced and the rate at which process heat is supplied in this mode.

(b) Determine the power produced and the rate of process heat supplied if only 60 percent of the steam is routed to the process heater and the remainder is expanded to the condenser pressure.

10–70 Consider a cogeneration power plant modified with regeneration. Steam enters the turbine at 6 MPa and 450°C and expands to a pressure of 0.4 MPa. At this pressure, 60 percent of the steam is extracted from the turbine, and the remainder expands to 10 kPa. Part of the extracted steam is used to heat the feedwater in an open feedwater heater. The rest of the extracted steam is used for process heating and leaves the process heater as a saturated liquid at 0.4 MPa. It is subsequently mixed with the feedwater leaving the feedwater heater, and the mixture is pumped to the boiler pressure. Assuming the turbines and the pumps to be isentropic, show the cycle on a T - s diagram with respect to saturation lines, and determine the mass flow rate of steam through the boiler for a net power output of 15 MW. *Answer: 17.7 kg/s*

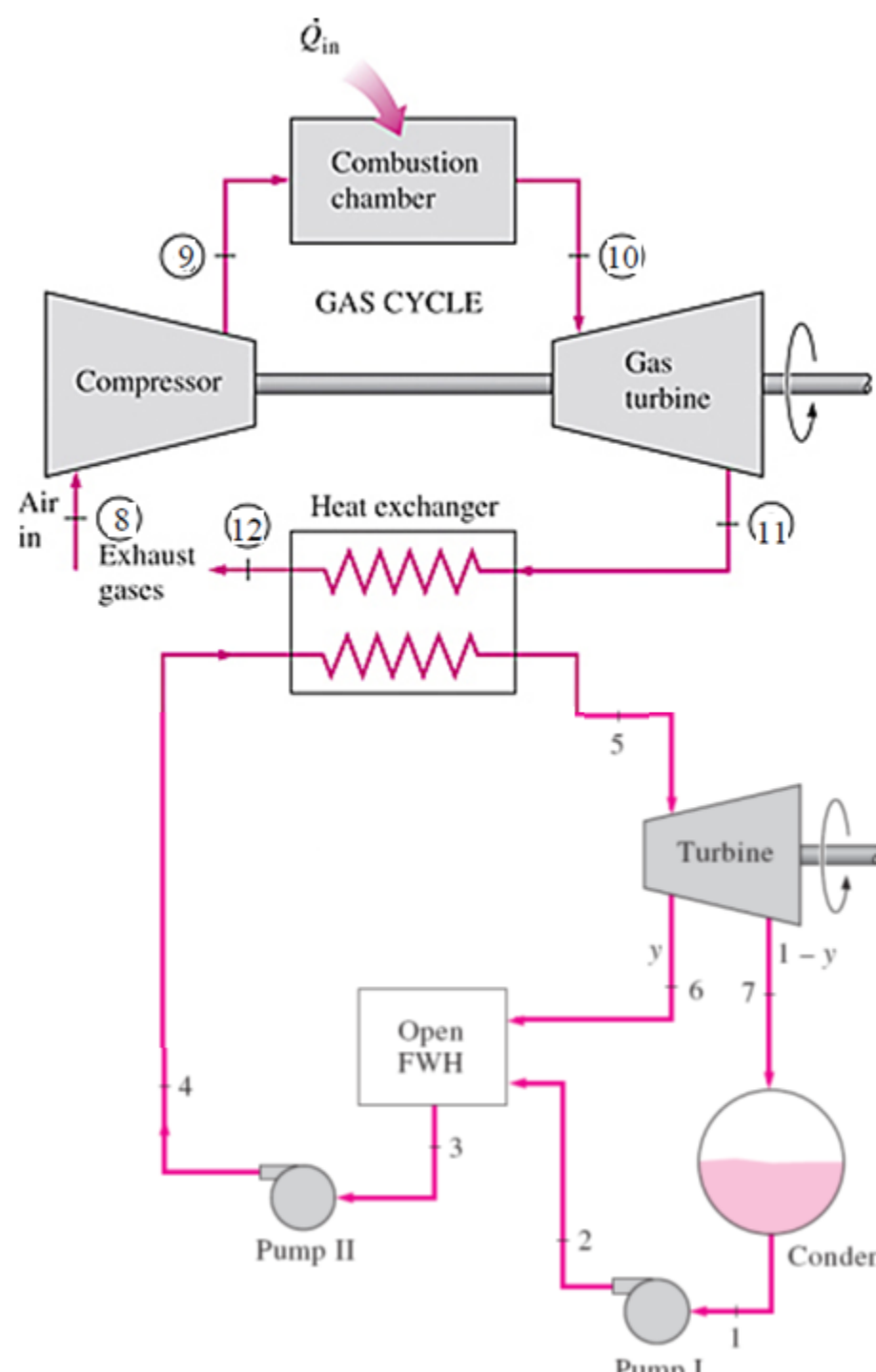


10–73 A cogeneration plant is to generate power and 8600 kJ/s of process heat. Consider an ideal cogeneration steam plant. Steam enters the turbine from the boiler at 7 MPa and 500°C. One-fourth of the steam is extracted from the turbine at 600-kPa pressure for process heating. The remainder of the steam continues to expand and exhausts to the condenser at 10 kPa. The steam extracted for the process heater is condensed in the heater and mixed with the feedwater at 600 kPa. The mixture is pumped to the boiler pressure of 7 MPa. Show the cycle on a T - s diagram with respect to saturation lines, and determine (a) the mass flow rate of steam that must be supplied by the boiler, (b) the net power produced by the plant, and (c) the utilization factor.



Combined Gas–Vapor Power Cycles

10–77  Consider a combined gas–steam power plant that has a net power output of 450 MW. The pressure ratio of the gas-turbine cycle is 14. Air enters the compressor at 300 K and the turbine at 1400 K. The combustion gases leaving the gas turbine are used to heat the steam at 8 MPa to 400°C in a heat exchanger. The combustion gases leave the heat exchanger at 460 K. An open feedwater heater incorporated with the steam cycle operates at a pressure of 0.6 MPa. The condenser pressure is 20 kPa. Assuming all the compression and expansion processes to be isentropic, determine (a) the mass flow rate ratio of air to steam, (b) the required rate of heat input in the combustion chamber, and (c) the thermal efficiency of the combined cycle.



Analysis (a) The analysis of gas cycle yields (Table A-17)

$$T_8 = 300 \text{ K} \longrightarrow h_8 = 300.19 \text{ kJ/kg}$$

$$P_{r_8} = 1.386$$

$$P_{r_9} = \frac{P_9}{P_8} P_{r_8} = (14)(1.386) = 19.40 \longrightarrow h_9 = 635.5 \text{ kJ/kg}$$

$$T_{10} = 1400 \text{ K} \longrightarrow h_{10} = 1515.42 \text{ kJ/kg}$$

$$P_{r_{10}} = 450.5$$

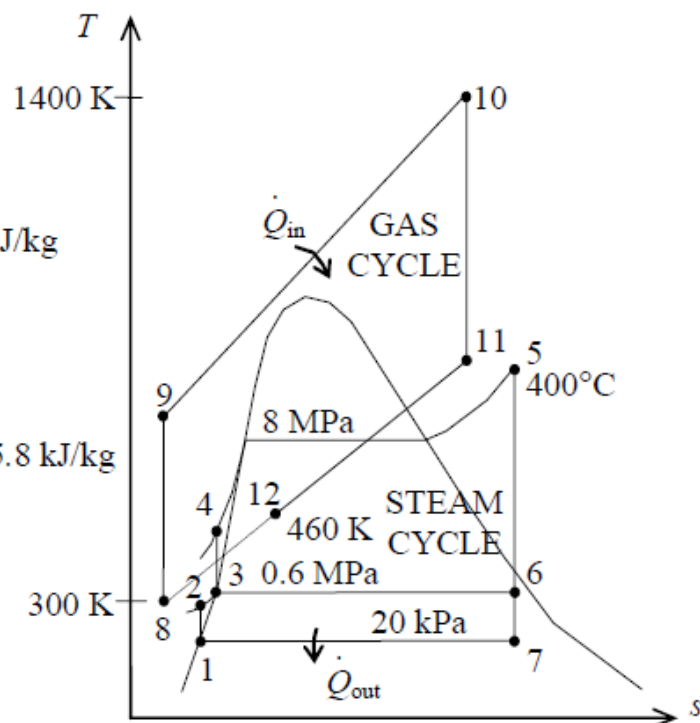
$$P_{r_{11}} = \frac{P_{11}}{P_{10}} P_{r_{10}} = \left(\frac{1}{14} \right) (450.5) = 32.18 \longrightarrow h_{11} = 735.8 \text{ kJ/kg}$$

$$T_{12} = 460 \text{ K} \longrightarrow h_{12} = 462.02 \text{ kJ/kg}$$

From the steam tables (Tables A-4, A-5, A-6),

$$h_1 = h_f @ 20 \text{ kPa} = 251.42 \text{ kJ/kg}$$

$$v_1 = v_f @ 20 \text{ kPa} = 0.001017 \text{ m}^3/\text{kg}$$



$$\begin{aligned}
 w_{pI,in} &= \nu_1(P_2 - P_1) \\
 &= (0.001017 \text{ m}^3/\text{kg})(600 - 20 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\
 &= 0.59 \text{ kJ/kg}
 \end{aligned}$$

$$h_2 = h_1 + w_{pI,in} = 251.42 + 0.59 = 252.01 \text{ kJ/kg}$$

$$h_3 = h_f @ 0.6 \text{ MPa} = 670.38 \text{ kJ/kg}$$

$$\nu_3 = \nu_f @ 0.6 \text{ MPa} = 0.001101 \text{ m}^3/\text{kg}$$

$$\begin{aligned}
 w_{pII,in} &= \nu_3(P_4 - P_3) \\
 &= (0.001101 \text{ m}^3/\text{kg})(8,000 - 600 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\
 &= 8.15 \text{ kJ/kg}
 \end{aligned}$$

$$h_4 = h_3 + w_{pII,in} = 670.38 + 8.15 = 678.53 \text{ kJ/kg}$$

$$\begin{aligned}
 P_5 = 8 \text{ MPa} &\left\{ \begin{aligned} h_5 &= 3139.4 \text{ kJ/kg} \\ T_5 = 400^\circ\text{C} &\left\{ \begin{aligned} s_5 &= 6.3658 \text{ kJ/kg} \cdot \text{K} \end{aligned} \right. \end{aligned} \right.
 \end{aligned}$$

$$\begin{aligned}
 P_6 = 0.6 \text{ MPa} &\left\{ \begin{aligned} x_6 &= \frac{s_6 - s_f}{s_{fg}} = \frac{6.3658 - 1.9308}{4.8285} = 0.9185 \\ s_6 = s_5 &\left\{ \begin{aligned} h_6 &= h_f + x_6 h_{fg} = 670.38 + (0.9185)(2085.8) = 2586.1 \text{ kJ/kg} \end{aligned} \right. \end{aligned} \right.
 \end{aligned}$$

$$\begin{aligned}
 P_7 = 20 \text{ kPa} &\left\{ \begin{aligned} x_7 &= \frac{s_7 - s_f}{s_{fg}} = \frac{6.3658 - 0.8320}{7.0752} = 0.7821 \\ s_7 = s_5 &\left\{ \begin{aligned} h_7 &= h_f + x_7 h_{fg} = 251.42 + (0.7821)(2357.5) = 2095.2 \text{ kJ/kg} \end{aligned} \right. \end{aligned} \right.
 \end{aligned}$$

Noting that $\dot{Q} \cong \dot{W} \cong \Delta \text{ke} \cong \Delta \text{pe} \cong 0$ for the heat exchanger, the steady-flow energy balance equation yields



$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \overset{\text{0 (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e \longrightarrow \dot{m}_s (h_5 - h_4) = \dot{m}_{\text{air}} (h_{11} - h_{12})$$

$$\frac{\dot{m}_{\text{air}}}{\dot{m}_s} = \frac{h_5 - h_4}{h_{11} - h_{12}} = \frac{3139.4 - 678.53}{735.80 - 462.02} = \mathbf{8.99 \text{ kg air / kg steam}}$$

(b) Noting that $\dot{Q} \equiv \dot{W} \equiv \Delta ke \equiv \Delta pe \equiv 0$ for the open FWH, the steady-flow energy balance equation yields

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \overset{\text{0 (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\sum \dot{m}_i h_i = \sum \dot{m}_e h_e \longrightarrow \dot{m}_2 h_2 + \dot{m}_6 h_6 = \dot{m}_3 h_3 \longrightarrow y h_6 + (1 - y) h_2 = (1) h_3$$

Thus,

$$y = \frac{h_3 - h_2}{h_6 - h_2} = \frac{670.38 - 252.01}{2586.1 - 252.01} = 0.1792 \quad (\text{the fraction of steam extracted})$$

$$\begin{aligned} w_T &= h_5 - h_6 + (1 - y)(h_6 - h_7) \\ &= 3139.4 - 2586.1 + (1 - 0.1792)(2586.1 - 2095.2) = 956.23 \text{ kJ/kg} \end{aligned}$$

$$\begin{aligned} w_{\text{net, steam}} &= w_T - w_{p, \text{in}} = w_T - (1 - y)w_{p, I} - w_{p, II} \\ &= 956.23 - (1 - 0.1792)(0.59) - 8.15 = 948.56 \text{ kJ/kg} \end{aligned}$$

$$\begin{aligned} w_{\text{net, gas}} &= w_T - w_{C, \text{in}} = (h_{10} - h_{11}) - (h_9 - h_8) \\ &= 1515.42 - 735.8 - (635.5 - 300.19) = 444.3 \text{ kJ/kg} \end{aligned}$$



The net work output per unit mass of gas is

$$w_{\text{net}} = w_{\text{net,gas}} + \frac{1}{8.99} w_{\text{net,steam}} = 444.3 + \frac{1}{8.99}(948.56) = 549.8 \text{ kJ/kg}$$

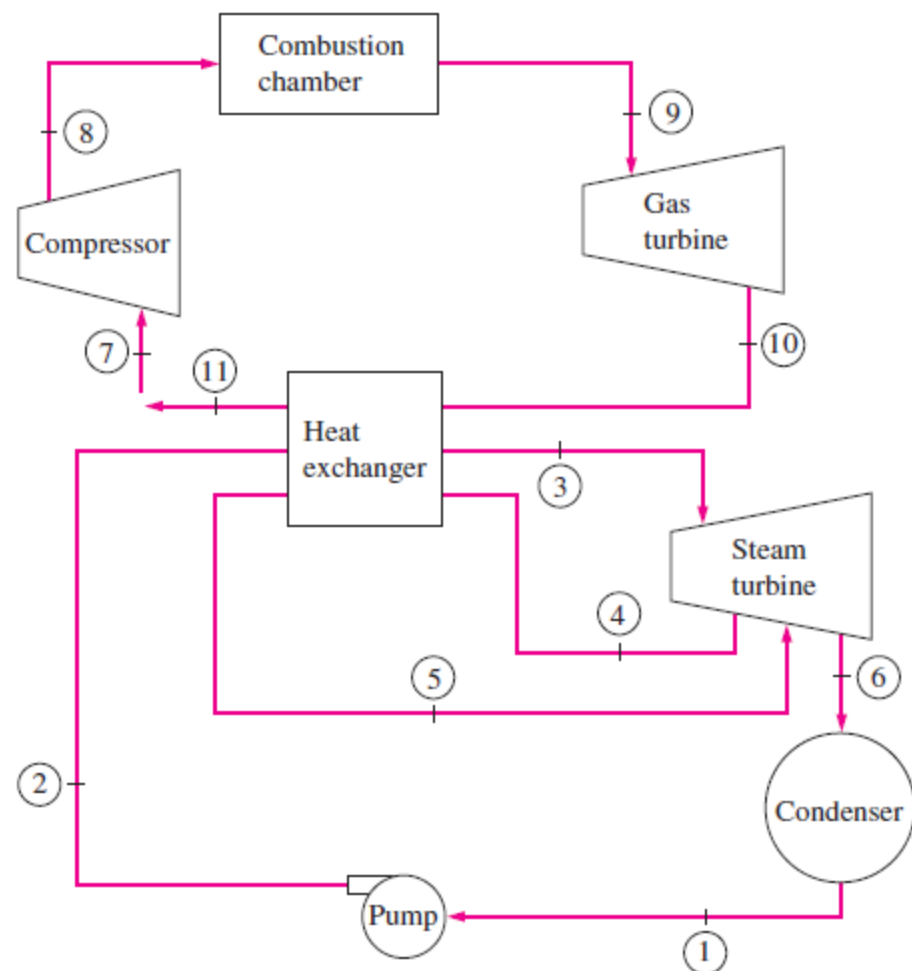
$$\dot{m}_{\text{air}} = \frac{\dot{W}_{\text{net}}}{w_{\text{net}}} = \frac{450,000 \text{ kJ/s}}{549.7 \text{ kJ/kg}} = 818.7 \text{ kg/s}$$

and

$$\dot{Q}_{\text{in}} = \dot{m}_{\text{air}}(h_{10} - h_9) = (818.5 \text{ kg/s})(1515.42 - 635.5) \text{ kJ/kg} = 720,215 \text{ kW}$$

$$(c) \quad \eta_{th} = \frac{\dot{W}_{\text{net}}}{\dot{Q}_{\text{in}}} = \frac{450,000 \text{ kW}}{720,215 \text{ kW}} = \mathbf{62.5\%}$$

10–81 Consider a combined gas–steam power cycle. The topping cycle is a simple Brayton cycle that has a pressure ratio of 7. Air enters the compressor at 15°C at a rate of 10 kg/s and the gas turbine at 950°C . The bottoming cycle is a reheat Rankine cycle between the pressure limits of 6 MPa and 10 kPa . Steam is heated in a heat exchanger at a rate of 1.15 kg/s by the exhaust gases leaving the gas turbine and the exhaust gases leave the heat exchanger at 200°C . Steam leaves the high-pressure turbine at 1.0 MPa and is reheated to 400°C in the heat exchanger before it expands in the low-pressure turbine. Assuming 80 percent isentropic efficiency for all pumps and turbine, determine (a) the moisture content at the exit of the low-pressure turbine, (b) the steam temperature at the inlet of the high-pressure turbine, (c) the net power output and the thermal efficiency of the combined plant.



Chapter 10

End of the Problems